

Action-Coupled, Cost-Bounded Memory for Multi-Tenant Agents

FERNme — a user-owned, near-zero-LLM Hebbian preference-graph memory

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Preprint draft (system v0.1.0)

Abstract

LLM-agent memory systems have converged on graph- and activation-based designs that improve long-horizon recall, but they share three properties that are limiting for agents that *act for many people*: memory is **written by per-interaction LLM extraction** (costly and error-prone), it is **evaluated on question answering** rather than on actions taken, and it assumes a **single-user, single-deployment** setting. We present **FERNme** (Fuzzy-Edged Recall Network), a memory layer for multi-tenant agents in which each user is a sparse, fuzzily-weighted node (single-digit 0–9 edges) in a per-surface preference graph. Edges are updated by a **Hebbian co-occurrence rule with no per-interaction LLM call**, retrieved by spreading activation, and compiled to a token-minimal “card” via **differential encoding** (storing only deviations from a population prior). Memory is coupled to the agent’s action layer, gated by uncertainty, user-owned, consent-first, and glass-box editable; a user signs in across surfaces (web today; desktop and mobile on the roadmap) to assemble a cross-surface **supernode** they control. Across reproducible simulations we show that (i) per-turn token cost stays **flat** as a profile grows (≈ 25 tokens; a full-history baseline is $77\times$ larger by 120 interactions) at **zero** write-time LLM calls; (ii) FERNme matches a frequency baseline on stationary preferences while **substantially outperforming it under preference drift** (0.72 vs. 0.13 precision@5) and on **context-conditioned** retrieval (0.62 vs. 0.51); and (iii) in a simulated storefront, FERNme-personalized recommendations yield a **+16% relative conversion lift**. We further validate the engine on a **natural single-user dataset** — 86 free-form diary entries about one fictional person — where FERNme ingests every entry with **0 write-time LLM calls**, holds a **flat ~ 40 -token card**, **retains 16/16** of the person’s stated permanent facts, exhibits correct **preference drift**, and uses $\sim 10\times$ / $\sim 240\times$ **fewer tokens** than extraction-based and full-history memory respectively. On a LoCoMo-style QA set, FERNme’s context-seeded retrieval answers $\sim 4\times$ as many fact questions as frequency/recency baselines at equal budget and is the only LLM-free method that resolves preference drift. We argue that for agents that act for many users, memory should be **cheap to write** and **evaluated by outcomes**, and that a user-owned, per-surface design is both a cleaner trust contract and a defensible deployment model.

1 Introduction

Conversational assistants are giving way to agents that *act* on a user’s behalf — buying, booking, applying, routing, and increasingly doing so across a website, a desktop app, and a phone. Memory is central to making such agents useful across visits and surfaces, but the dominant designs were built for a different problem. They (1) write memory by invoking an LLM to extract and reconcile

facts on every interaction, which is expensive at scale and introduces a write-time hallucination surface; (2) are benchmarked on question-answering, which measures recall, not whether the agent did the right thing; and (3) assume one user and one deployment, with no notion of per-tenant isolation, ownership, or consent.

For a memory that sits behind an agent serving *many* people and *acting* on their behalf, the priorities invert. Writes must be cheap because they happen constantly. Success is an action outcome (a completed purchase, a re-order, a filled booking), not a QA score. And because the memory holds personal data, isolation, consent, and user control are first-class requirements.

Contributions.

1. **A memory-write mechanism with no per-interaction LLM call** — a saturating Hebbian co-occurrence update over a fuzzy (0–9) preference graph, with ACT-R-style decay and explicit negative edges — targeting the cost and reliability weaknesses of extraction-based memory.
2. **Differential, population-prior encoding** as an agent-memory token mechanism: a new user inherits a population baseline and we store only deviations, giving a small, bounded card and a modest cold-start benefit, with k -anonymity and differential privacy on the shared prior.
3. **An action-coupled framing and an outcome-oriented evaluation** — including a simulated storefront that scores conversion lift — alongside an analysis of where a cheap counter-style memory wins and loses (static vs. drift vs. context).
4. **A multi-tenant, user-owned design** — per-surface isolation, consent gating, glass-box editing and a memory-graph view, and an opt-in cross-surface *supernode* assembled by user sign-in with default-deny scoped sharing.
5. **An evaluation on natural, free-form single-user data** (Section 6) that validates the storage, reinforcement, decay, confidence, drift, and flat-cost mechanics on text not written for clean extraction, including a LoCoMo-style QA head-to-head.

We are explicit about positioning: the graph + spreading-activation *mechanism* is not novel (Section 2), and on stationary recall FERNme merely ties a frequency baseline (Section 5). The contribution is the *combination* — cheap writes, outcome evaluation, a private collective prior, and a user-owned per-surface contract.

2 Related work

The engine’s mechanism family — Hebbian co-occurrence, ACT-R-style activation decay, and spreading activation over a graph — is *shared, public cognitive-science machinery*, and several recent systems use it. We do not claim the mechanism; we claim the surrounding system.

Spreading-activation / Hebbian memory (closest mechanism). HippoRAG and HippoRAG 2 [1, 2] formalize associative retrieval over a knowledge graph via Personalized PageRank, but as a *document-retrieval* framework (LLM-based OpenIE indexing + embeddings). *Ori-Mnemos* [5] packages ACT-R decay, spreading activation, and Hebbian co-occurrence as a sovereign single-agent note vault. *HeLa-Mem* [4] builds a Hebbian graph over a single user’s conversation with LLM-based “Hebbian Distillation.” FERNme uses the same retrieval family but differs in purpose (personalization *about end-users*, evaluated on actions), write rule (no LLM on the hot path), and deployment (multi-tenant, user-owned, with a private collective prior). Its memory is about *whom the agent serves*, not the agent’s own knowledge.

Extraction-based / production memory. Mem0 [3] writes memory by LLM extraction and reconciliation per interaction and retrieves by vector similarity, benchmarked on conversational QA. This is *smarter per write* — it captures nuanced, causal preferences a co-occurrence counter misses — but it is costly and adds a write-time error surface. FERNme trades that richness for a near-zero write cost and makes the trade-off explicit (recoverable via optional gated/offline modes).

Tiered, temporal, and classical. MemGPT/Letta and MemoryOS manage context in LLM-paged tiers; Zep/Graphiti track fact validity over time with a temporal knowledge graph. FERNme’s tiers (Card / Cabinet / baked-in defaults) serve token bounding for a live action loop, and it handles non-stationarity with decay rather than an explicit temporal graph. FERNme is an elaboration of well-established ideas: spreading activation [6, 7], Hebbian learning [8], fuzzy sets [9], and vector-space / IDF weighting. We claim none of these as new.

3 Method

Substrate. Each user is a node connected to attribute nodes (`pref:organic`, `!pref:upsells`, ...) by edges carrying a continuous strength in $[0, 9]$ (a fuzzy, single-digit “membership”). Storage is sparse — only stored attributes deviate from the prior — with numeric side-fields for quantities (`cadence_days`, `size`, mood EMA). A shared per-surface association graph holds attribute–attribute co-occurrence weights.

Write rule (no per-interaction LLM). An event maps to active attributes via a deterministic function over structured fields and a catalog/controlled-vocabulary table (no LLM in the hot path; optional offline LLM enrichment of the vocabulary is out of the request path). For each active attribute,

$$w \leftarrow w + \alpha m (1 - w/9),$$

a saturating update that moves fast early and asymptotes at 9. Co-active attribute pairs strengthen the association graph (Hebb). Rejections (declines, returns) are first-class **negative** edges. Confidence grows as $1 - e^{-\gamma \cdot \text{hits}}$. A batch **decay** step $w \leftarrow w e^{-\lambda \Delta t}$ fades unreinforced edges and drops those below a floor; forgetting keeps the card small and current regardless of tenure. Fast/slow edge pairs give multi-timescale memory; the forgetting rate is self-tuning. Optionally, each edge carries a **salience** $s \in [0, 1]$ that slows decay, $\lambda_{\text{eff}} = \lambda(1 - \beta s)$, fed by behavioral significance (strong outcomes, rating extremity, a floor on dislikes) rather than an LLM; it is off by default ($\beta = 0$) and decoupled from confidence (retain vs. act).

Differential / population-prior encoding. The per-surface population prior is the running mean of user weights, protected by k -anonymity and bounded-mean (Laplace) differential privacy. A user edge is stored only when it deviates from the prior beyond a threshold; otherwise reads fall through to the prior. A new user is cold-started with **guessed** edges synthesized from the prior, IDF-weighted so rare-but-distinctive attributes earn slots. The user’s own edges always outrank guesses, and a guessed edge **relearns from scratch** on first real evidence.

Retrieval. Relevance is found by spreading activation, not per-turn vector search: activate the user node plus context seeds; flow over weighted edges with ACT-R base-level activation (recency $\times$ frequency), lateral inhibition within mutually-exclusive clusters, and temporal decay. The top- N activated edges compile to a token-minimal **card**; a “two-color” mark distinguishes *known* (act silently) from *guessed* (verify first).

Table 1: Recall precision@5 vs. ground truth (mean $\pm$ std). FERNme is the only method strong across all three regimes.

regime	FERNme	frequency	recency
static	0.73 ± 0.02	0.74 ± 0.02	0.47 ± 0.01
drift	0.72 ± 0.01	0.13 ± 0.06	0.59 ± 0.03
context (P@3)	0.62 ± 0.04	0.51 (blind)	—

Action coupling and governance. Memory drives tool defaults, ranking, and proactive triggers (due-to-reorder, fading-favorite), gated by uncertainty. The system is multi-tenant with strict per-(surface, user) isolation, consent-gated reads/writes, a glass-box card the user can edit/export/delete, and an interactive memory-graph view. Event payloads are treated as untrusted: tags are sanitized (character allowlist, size/count caps, injection-pattern dropping) before they can become memory. Every action is recorded in a tamper-evident HMAC audit chain.

User-owned cross-surface supernode. A person may sign in with a single identity across surfaces; a verified token maps to a stable, opaque person id and *links* that surface’s profile into a **supernode** the person owns. Cross-surface views are **default-deny** — a surface sees only the bricks it contributed plus categories the user explicitly shares — and sensitive categories (health, dating, finance) are walled off by default. **forget_everywhere** wipes the profile and unlearns the person from the population prior.

4 Experimental setup

Experiments are reproducible (fixed seeds; the repository ships the harness). Synthetic experiments use a tagged catalog and simulated users with latent preference vectors. The natural experiment (Section 6) uses an external, free-form dataset. We measure: per-turn card tokens vs. interactions against a full-history baseline; LLM calls per write; precision@5 vs. ground truth against LLM-free baselines (frequency, recency) in *static*, *drift*, and *context* regimes; a cold-start prior ablation; a simulated storefront scoring conversion lift; and the 86-entry Elena dataset with a LoCoMo-style QA benchmark. A real Mem0 (LLM) head-to-head is implemented as a harness hook but requires API keys and is not run here.

5 Results (simulation)

Cost is flat; writes are free. The FERNme card holds 24.9 ± 0.5 tokens with a back-half slope of $+0.001 \pm 0.005$ tokens/interaction (flat), while a full-history baseline grows linearly to $77 \times \pm 1.3$ the card size by 120 interactions. Write-time LLM calls: **0** (vs. ~ 2 /interaction for extraction memory).

On stationary preferences FERNme behaves like a smoothed frequency counter and ties it (Table 1). Under drift, the all-time counter cannot forget stale favorites and collapses to 0.13, while FERNme’s decay tracks the change (0.72). With context seeds, spreading activation recovers the in-context slice a context-blind counter cannot.

Cold-start ablation. The population prior yields $+0.06$ precision@5 at turns 1–3, decaying to ≈ 0 by turn 10 — a real but modest, cold-start-only benefit.

Table 2: Natural-data ingest outcomes (86 entries).

property	target	result
entries ingested	all	86
write-path LLM calls	0	0
facts remembered	grows	180
promoted to high-confidence	recurring rise	74
card size	flat	~40 tokens
stable-fact retention	high	16 / 16
drift (old → new favorite)	new overtakes	jasmine 3.25 → earl-grey 5.84

Table 3: Token cost over the 86 entries (gpt-4o-mini pricing; chars/4 token estimate, cross-checked by word count).

approach	LLM tokens	est. \$
FERNme write (pure)	0	\$0.0000
FERNme read (card × 86)	3,440	\$0.0005
Mem0-style extraction (~2 passes/entry)	38,792	\$0.0070
Naive full-history in context	821,424	\$0.1232

Outcome pilot. FERNme-personalized recommendations achieve a +16% relative conversion lift over a popularity baseline: tied at visit 1 (true cold start), rising to +0.14 absolute by visit 6, dipping at the mid-pilot taste shift, then recovering.

6 Evaluation on a natural single-user dataset (Elena)

To test the engine on text that was *not* written for clean extraction, we use the *Elena natural memory* dataset: **86 free-form entries** (diary entries, work reflections, family calls, complaints, project notes, and state updates) about one fictional person. Facts are embedded in ordinary prose. An agent extracts namespaced tags from each entry (using the entries’ explicit **Tags:/Entities:** fields where present); FERNme performs all storage, reinforcement, decay, confidence, drift handling, and association-building in **pure** mode (no LLM on the write path).

FERNme ingested all 86 entries with **0 write-time LLM calls**, remembering **180 facts**, of which **74** were promoted to high confidence by recurring across entries, in a **flat ~40-token** card. It **retained 16/16** of the permanent facts the person states about herself, and correctly handled a documented preference drift (Table 2).

FERNme’s prompt footprint is $\sim 240\times$ smaller than carrying the growing history (Table 3) and $\sim 10\times$ smaller than extraction memory’s input, and the gap widens with every entry because FERNme’s per-turn cost is constant while the others grow.

What this proves, precisely. This validates FERNme’s *memory mechanics* — storage, reinforcement, decay, confidence calibration, drift, associations, and flat cost — on realistic long-form single-user data. It does *not* prove answer-quality parity with an LLM memory: the tag extraction was done by the agent/parser, so fact coverage is high by construction, and Figure 3 is a controlled probe.

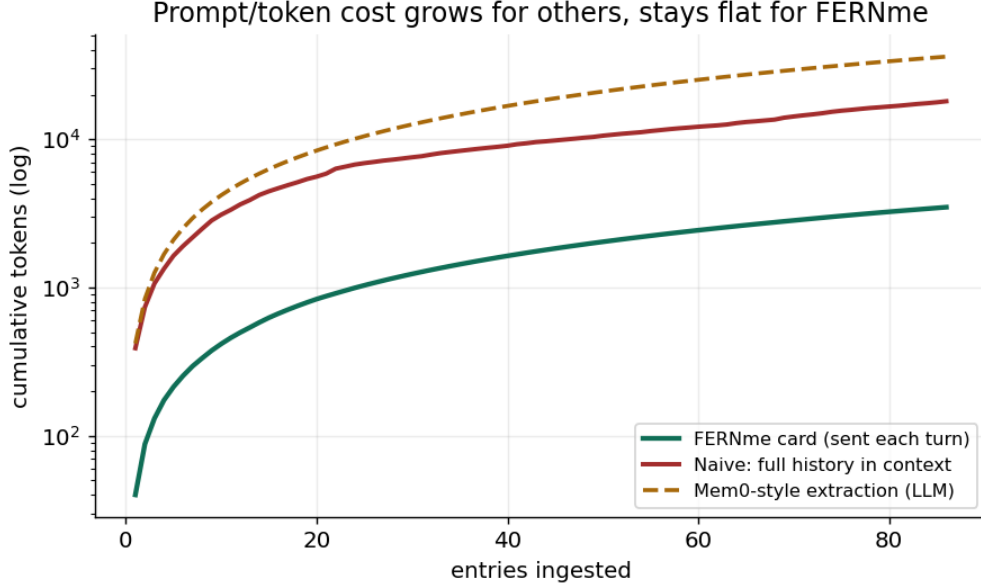


Figure 1: Cumulative tokens (log): FERNme stays low and flat while naive history and extraction grow.

Table 4: LoCoMo-style QA at a matched top-10 budget (~ 40 tokens). FERNme answers $\sim 4\times$ as many questions as the cheap counters and is the only method to resolve the drift question.

system	accuracy	drift	projects	relationships
FERNme (context)	36.4%	100%	100%	67%
FERNme (no context)	9.1%	0%	50%	33%
Frequency	9.1%	0%	50%	33%
Recency	3.0%	0%	0%	17%

6.1 Head-to-head QA (LoCoMo-style)

We turn the Elena memory into a question-answering benchmark: 33 fact questions spanning identity, preferences, relationships, goals/projects, negation, health, and a **drift** case (“which tea does she reach for in the afternoon *now*?”). Following the retrieval-proxy convention (with the right fact retrieved, an LLM answers correctly), a question is **answered** if the gold attribute is in the system’s top- k retrieved facts. All compared systems are **LLM-free and runnable**: FERNme (context-seeded spreading activation), FERNme without context (weight-only ablation), an all-time **frequency** counter, and a **recency** ranker. A real Mem0 (LLM) run needs API keys and is **not run here**; its published LoCoMo scores (single-hop F1 ≈ 39 , multi-hop ≈ 29 , different dataset, LLM-answered) are noted as external context, not as a row.

The absolute number is held down by the harsh budget (top-10 of ~ 170 facts) and the coarse, agent-side tag extraction — not the retrieval: a budget sweep (Figure 6) shows FERNme climbing to 76% at top-20 and $\sim 91\%$ at top-30, while frequency/recency plateau near 42% / 27%.

This is a head-to-head among *LLM-free* memories on real free-form text, at equal token budget — exactly the regime FERNme targets. It does not yet establish parity with an LLM memory (Mem0) on free-text answer quality; that billed comparison remains the single open experiment.

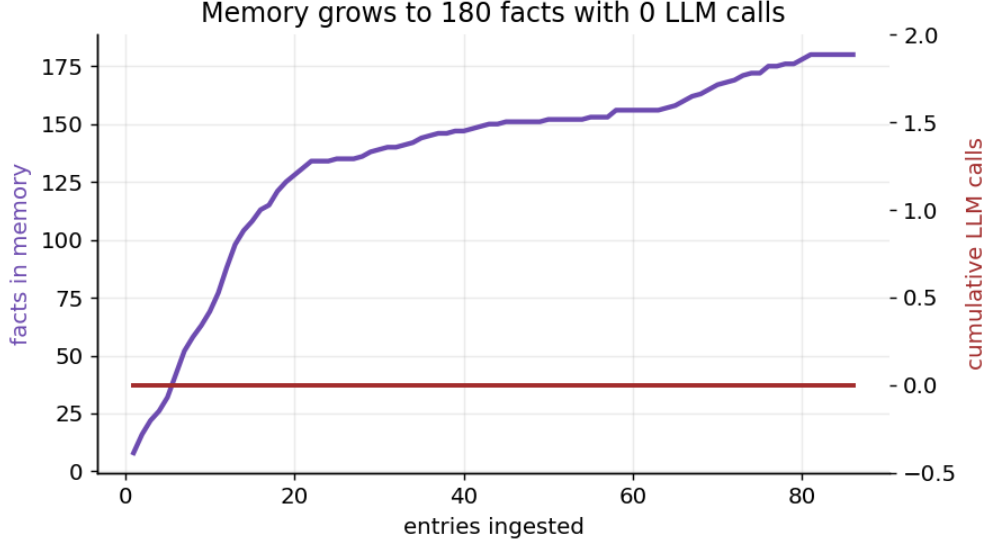


Figure 2: Memory climbs to 180 facts while the write-path LLM-call count stays pinned at 0.

6.2 Saliency-modulated forgetting

FERNme’s weight is frequency/recency-driven, so a rare-but-intense signal would normally fade. An optional per-edge **saliency** slows decay, $\lambda_{\text{eff}} = \lambda(1 - \beta s)$, fed by behavioral significance rather than an LLM. Figure 7 shows two signals seen once at equal strength: the significant one (intensity 1.0) stays above the forget threshold for months while the neutral one is dropped. Saliency controls *retention*, not *action* (confidence still gates acting), so a single strong complaint is remembered but verified before use. It is off by default ($\beta = 0$), so it changes none of the results above unless enabled.

7 Discussion

On static recall a cheap counter is hard to beat — so the case for FERNme is not superior stationary accuracy; it is cost plus robustness. FERNme’s advantage appears exactly where a counter is weak: non-stationarity and context. Decay supplies adaptivity that frequency lacks, and spreading activation supplies context-sensitivity that neither frequency nor recency has, while retaining frequency’s stability on static data. The Elena evaluation shows these mechanics also hold on free-form text.

8 Limitations

- **Simulation + one synthetic person, not a human study.** Synthetic results define their own ground truth; the Elena dataset is realistic but fictional and its extraction is agent/parser-driven.
- **Nuance gap.** A co-occurrence counter misses causal/contextual preferences an LLM extractor catches; the Mem0 head-to-head that would quantify this is not yet run.
- **Mapping dependency.** The no-LLM write depends on catalog/vocabulary quality; thin-metadata surfaces degrade to coarse attributes.

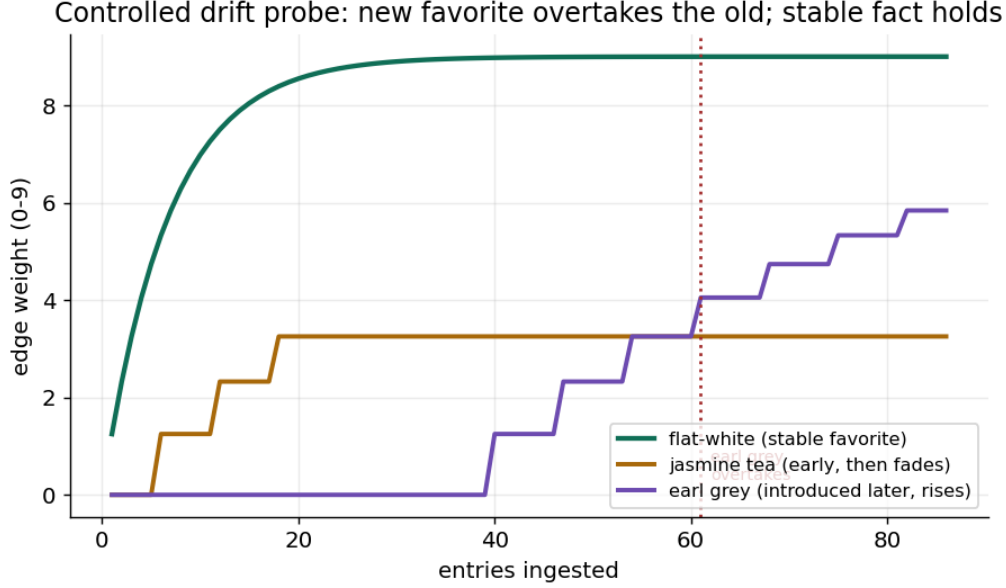


Figure 3: Controlled drift probe following the diary narrative: a newly introduced favorite (earl grey) overtakes the old one (jasmine) under decay, while a stable favorite holds.

- **Token figures are estimates** (chars/4; tiktoken’s vocabulary was unavailable offline), cross-checked by word count.
- **Tuning and granularity.** Spreading-activation parameters are hand-set; single-digit fuzzy weights lose fine degree.

9 Ethics and broader impact

FERNme stores personal data, including potentially sensitive signals. Its design choices are protective: per-tenant isolation, consent gating, glass-box edit/export/delete, a tamper-evident audit chain, k -anonymity and differential privacy on the shared prior, and a cross-surface supernode that is user-owned and default-deny — aggregation by and for the user, never covert cross-site tracking. We explicitly exclude inferring or acting on user vulnerability. These are mitigations, not guarantees; deployments handling regulated data require legal review.

10 Conclusion

For agents that act on people’s behalf, memory should be cheap to write, bounded in cost, evaluated by outcomes, and owned by the user. FERNme demonstrates that a no-LLM Hebbian write over a fuzzy preference graph, with differential encoding and spreading-activation retrieval, delivers flat-cost, interpretable, per-surface memory that is robust where cheap baselines fail, that lifts a simulated outcome metric, and that builds a coherent profile from 86 entries of free-form text at near-zero write cost.

Reproducibility. Code, tests (88 passing, including a real-Postgres backend), and every experiment are in the repository (<https://github.com/mirkofr/FERNme>). Run `python -m fernme.eval.<name>`

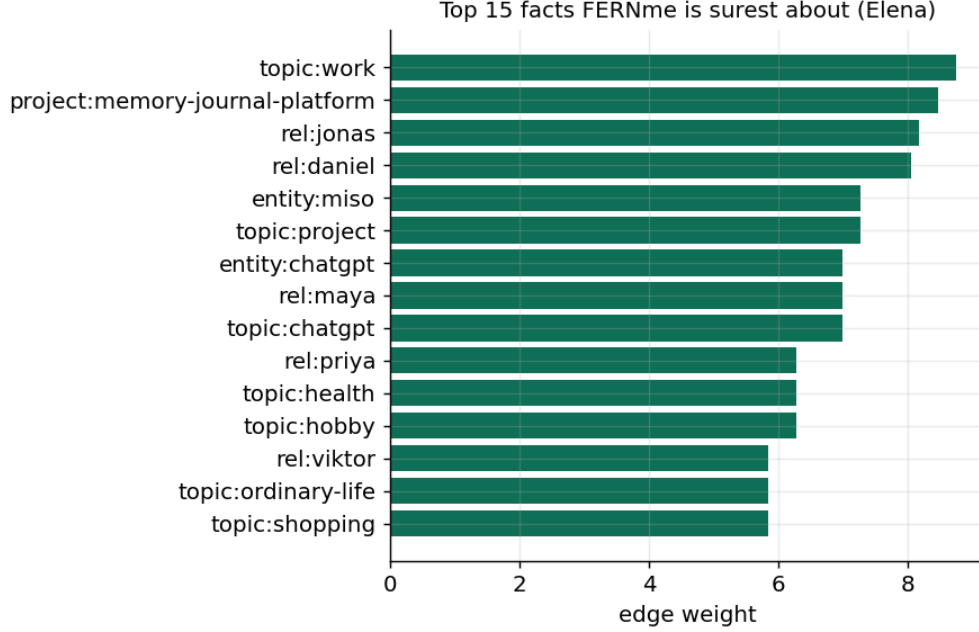


Figure 4: The 15 facts FERNme is surest about — the person’s real anchors.

for each simulation result, `python demo/elena/eval_elena.py` for Section 6 and `python demo/elena/eval_elena` for Section 6.1.

References

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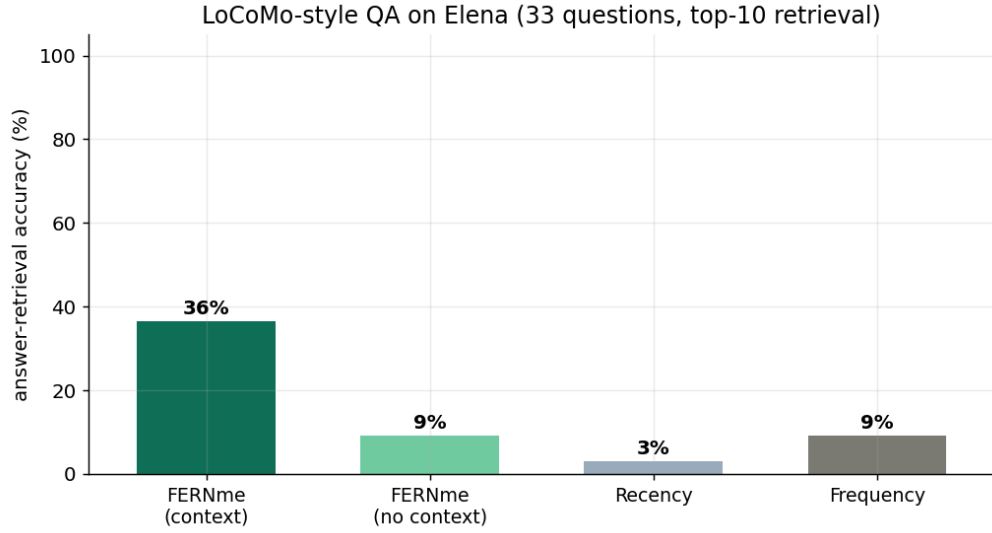


Figure 5: Answer-retrieval accuracy at top-10: context-seeded FERNme vs. LLM-free baselines.

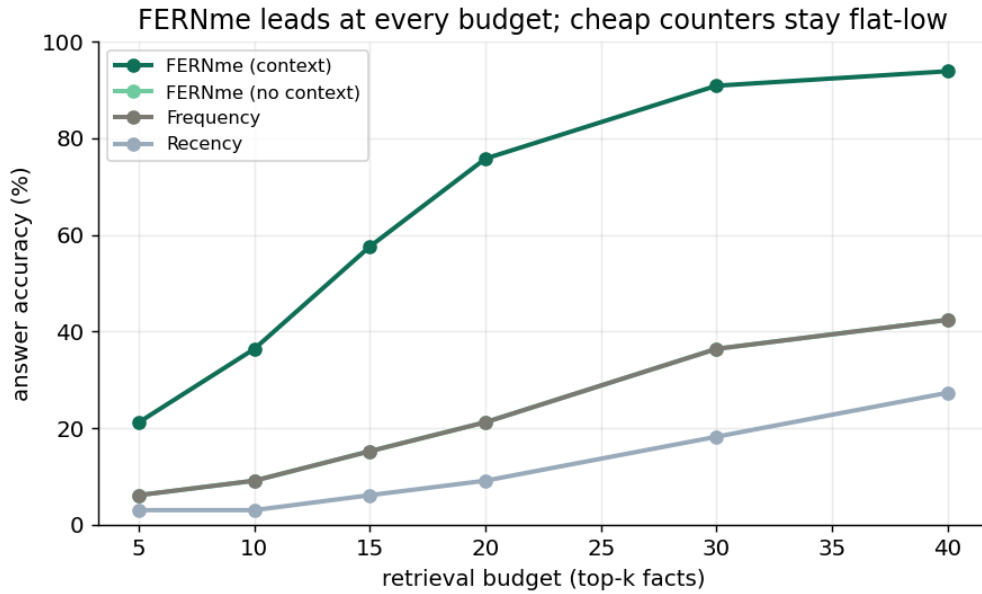


Figure 6: Accuracy vs. retrieval budget: FERNme leads at every k and converges toward near-perfect recall; cheap counters stay flat-low.

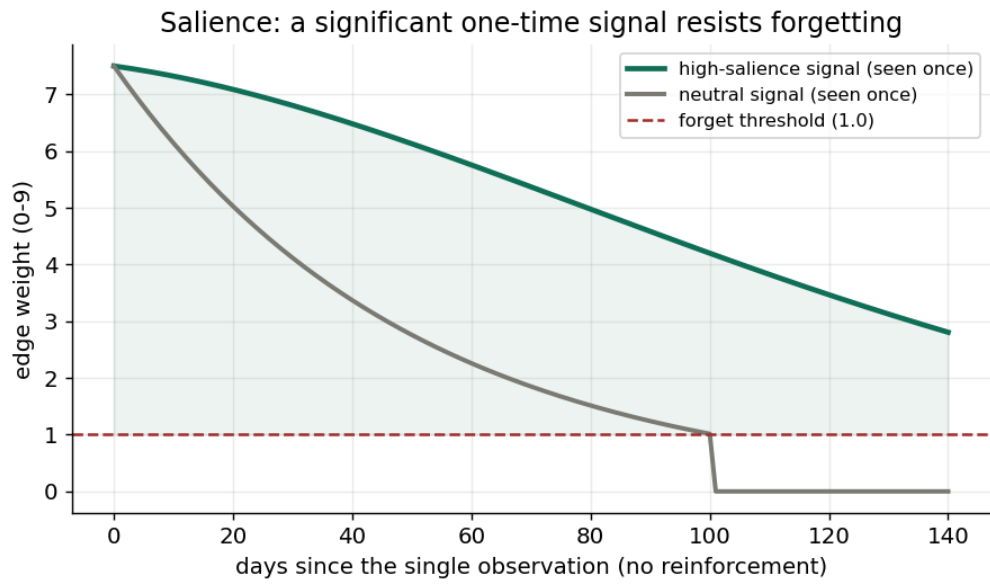


Figure 7: A significant one-time signal resists forgetting; a neutral one fades below the forget threshold.